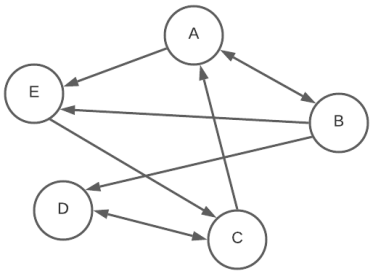
American Computer Science League

2021-2022 ● Contest 4: Short Problems Solutions ● Classroom Division

1. Graph Theory



The directed graph is:

$$\begin{vmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{vmatrix}$$

Adding the entries in the squared matrix gives the paths of length 2: 15.

D. 15

2. Graph Theory

There are 4: ABA, ABCA, ADA, and ADCA.

C. 4

3. Digital Electronics

The digital circuit translates to: $A\ (\overline{A\ B\ +\ C})$

$$\begin{aligned} A\ (\overline{A\ B\ +\ C}) &= A\ \overline{A}\ \overline{B}\ \overline{C} \\ &= A\ (\overline{A}\ +\ \overline{B})\ \overline{C} \\ &= A\ \overline{A}\ \overline{C}\ +\ A\ \overline{B}\ \overline{C} \\ &= A\ \overline{B}\ \overline{C} \end{aligned}$$

This is TRUE for only (1, 0, 0).

B. 1

4. Digital Electronics

The Boolean expression for this circuit is: $\overline{(A + \overline{A}\,B)\,B}$

$$\begin{aligned}\overline{(A + \overline{A}\,B)\,B} &= \overline{A + \overline{A}\,B} + \overline{B}\\ &= \overline{A}\,\overline{\overline{A}\,B} + \overline{B}\\ &= \overline{A}(A + \overline{B}) + \overline{B}\\ &= \overline{A}\,A + \overline{A}\,\overline{B} + \overline{B}\\ &= \overline{A}\,\overline{B} + \overline{B}\\ &= \overline{B}\,(\overline{A} + 1)\\ &= \overline{B}\end{aligned}$$

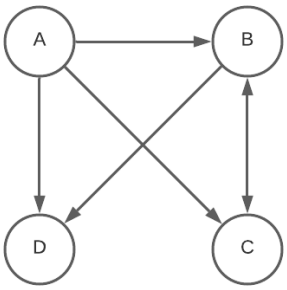
E. $\overline{B}$

5. Assembly Language

The program calculates 5! = 5*4*3*2*1 = 120

D. 120

6. Graph Theory



The original directed graph is:

0	1	1	1	2		0	1	1	1
0	0	1	1			0	1	0	0
0	1	0	0		=	0	0	1	1
0	0	0	0			0	0	0	0

This graph has 6 paths of length 2.

Now add directed edge DC.

B. 3

0	1	1	1	2		0	1	2	1
0	0	1	1			0	1	1	0
0	1	0	0		=	0	0	1	1
0	0	1	0			0	1	0	0

This graph has 9 paths of length 2 (3 more).

7. Graph Theory

The cycles from A are : AFCDA, AFEBA, AFEBCDA, AFEDA, ADA
There are 3 cycles of length 4: AFCDA, AFEBA, and AFEDA.

C. 3

8. Digital Electronics

The Boolean expression represented by the following digital circuit is:

$$\overline{A + AB}\,(\overline{BC} + C) = (\overline{A}\,\overline{AB})\,(\overline{B} + \overline{C} + C)$$
$$= \overline{A}\,(\overline{A} + \overline{B})\,1$$
$$= \overline{A} + \overline{A}\,\overline{B}$$
$$= \overline{A}\,(1 + \overline{B})$$
$$= \overline{A}$$

A. $\overline{A}$

9. Digital Electronics

The Boolean expression for the digital circuit is:

$$(\overline{AB} \oplus (B + C))\,C = (\overline{\overline{AB}}\,(B + C) + \overline{AB}\,\overline{B + C})\,C$$
$$= (AB + ABC + \overline{(A + B)}\,\overline{B}\,\overline{C})\,C$$
$$= (AB + \overline{A}\,\overline{B}\,\overline{B}\,\overline{C})\,C$$
$$= ABC$$

This is TRUE for only (1, 1, 1)

B. 1

10. Assembly Language This program prints the factors of 12 (1, 2, 3, 4, 6, 12). There are 6 of them.	E. 6